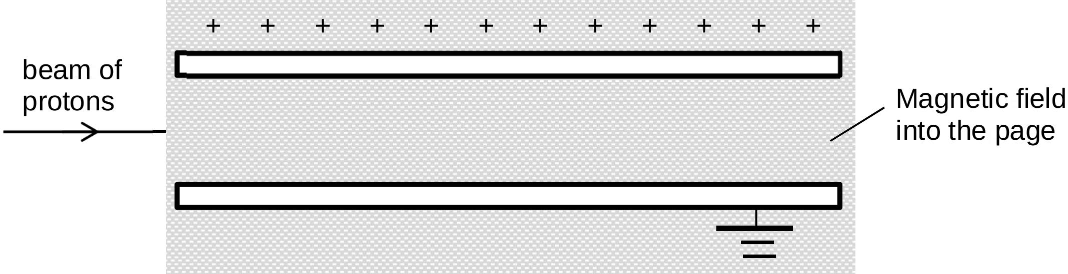
22	A velocity selector for protons has an electric field of strength 3.0×10^{5} V m^{-1}, produced by two horizontal plates. The upper plate is connected to a positive potential and the lower plate is earthed. A magnetic field of flux density 1.5×10^{-2} T, directed into the plane of the paper, is at right-angles to the electric field as shown in the diagram.



Which of the following could be the possible speed of a proton and its observation?

	speed	observation
A	1.8×10^7 m s^{-1}	undeflected
B	2.0×10^7 m s^{-1}	deflected downwards
C	2.2×10^7 m s^{-1}	deflected downwards
D	2.4×10^7 m s^{-1}	deflected upwards

Answer: D

Protons entering the velocity have a range of speeds.

If protons are undeflected, Upwards magnetic force = Downwards electric force.

i.e. Bqv = qE

$v = E/B = 3.0 \times 10^5 / 1.5 \times 10^{-2}$

$= 2.0 \times 10^7$ m s^{-1}.

Protons with speeds higher than this value will have a greater magnetic force, but the electric force is constant regardless of speed, hence deflected upwards.

Conversely, protons with speeds lower than this value will deflect downwards.